

Jaypee University of Engineering and Technology, Guna
B.Tech. (Semester II)
Object Oriented Programming Lab (CS202)
Lab Exercise 2

1. Write a program to generate random numbers in given range $[n, m]$ by asking the values of n and m from user at runtime.

Test Case

Input: $m=10, n=50$

Output: 34

2. Write a program which finds a four-digit number AABB which is a perfect square. A and B represent different digits. For example, 7744 is a four-digit perfect square number which is also satisfying the condition AABB i.e. first two digits (AA=77) are same and last two digits (BB=44) are same
3. Write a function that finds the minimum and the maximum value in an array of N integers. Inputs to the function are the array of integers, an integer variable containing the length of the array and pointers to integer variables that will contain the minimum and the maximum values. The function prototype is:

void minmax(int array[], int length, int * min, int * max);

Write a main function that uses this function to find and display the minimum and the maximum values of an array of integers.

4. Write a program to calculate factorial of a number N through recursion.
5. Write a program which takes your full name as input and returns the length of your name without using any string library functions. [Note: Write the program by passing string into function and returning the length of string from the function]
6. Write a pointer version of the function *strcat(s,t)* which concatenates the string t to the end of string s .
7. Write the function *strend(s,t)*, which returns 1 if the string t occurs at the end of the string s , and zero otherwise.

Test Case1

Input:

$s = \text{"Object Oriented Programming using C++"}$
 $t = \text{"Using C++"}$

Output: 1

Test Case2

Input:

$s = \text{"Object Oriented Programming using C++"}$
 $t = \text{"Programming"}$

Output: 0